Rohit Kumar

AI Engineer · LLMs, Agentic AI & Computer Vision

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Summary

AI Engineer at Armada AI, building retrieval and evaluation systems. Sole owner of its Support RAG Agent and sole engineering owner of Almanac, its on-device document Q&A application. Previously worked on diffusion-based product imagery and novel-view synthesis at Avataar AI. M.Tech in AI from IISc Bangalore; research published at WACV 2025 and CVPR 2026 Workshop.

Skills

AI/ML: LLMs, RAG, Agentic AI, Computer Vision, Diffusion Models, NLP

Model development: PyTorch, Supervised fine-tuning (SFT), LoRA, QLoRA, Distributed training

Engineering: Python, SQL, LangGraph, Hybrid Search, Evals (RAGAS), FastAPI, Qdrant, Kubernetes, Docker, SLURM

Publications

AttriStory: Fine-Grained Attribute Realization for Visual Storytelling with Diffusion Models
Manogna Sreenivas*, **Rohit Kumar***, Soma Biswas · CVPR 2026 Workshop
Paper Project page

TACLE: Task and Class-aware Exemplar-free Semi-supervised Class Incremental Learning
Jayateja Kalla*, **Rohit Kumar***, Soma Biswas · WACV 2025
Paper Code Project page

* Equal contribution.

Invited reviewer: NeurIPS, CVPR & ECCV 2026.

Experience

AI Engineer · **Armada AI** Jun 2025 - Present

Trivandrum, India

Manager: Naveen Nair

- Built and deployed the Support RAG Agent for Armada's internal Customer Support team (sole owner). Sole engineering owner of Almanac, an ongoing application designed for air-gapped document Q&A
- Designed the Support Agent's CRAG + controllable ReAct workflow; shared reformulation and embedding across branches reduced cost per query/run to approximately one-third of the initial graph on similar workloads
- Built Support Agent evaluation at two levels: retrieval-confidence and CRAG-score gating before generation, and RAGAS-based response evaluation against curated ground truth
- Built an online-learning BM25 index with incremental update and unlearn, addressing organizational-vocabulary gaps that dense retrieval missed
- Replaced per-format ingestion pipelines with a normalized-Markdown intermediate representation shared across file formats
- Fine-tuned models for Almanac using SFT, LoRA and QLoRA

Experience (continued)

Research Engineer · **Avataar AI** Jul 2024 - Apr 2025

Bangalore, India

Managers: Sohil Shah, Shubham Goel

- Built a Flux and ControlNet pipeline for lifestyle product imagery; modified diffusion sampling followed by intrinsic decomposition for product reconstruction and relighting
- Continued an existing sparse-view-conditioned diffusion project, fine-tuning a Stable Diffusion model for novel-view synthesis with distributed training and SLURM
- Built classification for low-data scenarios with CLIP, BLIP2, and Qwen2.5

Teaching Assistant · **IISc Bangalore** Aug 2023 - Apr 2024

Bangalore, India

- Teaching assistant for Digital Image Processing and Signal System in Practice
- Worked on continual learning (L2P, DualPrompt), self-supervised learning (MoCo, SimCLR), frequency-domain filtering and image features

Projects

LLM Inference Planning & Optimization Case study

Built a planner to predict KV-cache requirements, latency, throughput and token costs before trial deployment. Selected hardware and serving configurations, then deployed inference workloads.

Document-model evaluation Case study

Compared table-structure, OCR, layout and embedding models; built interchangeable pipeline components for model selection.

Novel-view synthesis Case study

Continued an existing Zero123-like diffusion project at Avataar, with initial setup by a senior engineer. Fine-tuned sparse-image-conditioned models using distributed training and SLURM; used all 32 H100 GPUs for a few days.

Education

M.Tech in Artificial Intelligence 2022 - 2024

Indian Institute of Science (IISc), Bangalore · CGPA: 8.0/10.0

Thesis advisor: Prof. Soma Biswas

B.Tech in Electrical Engineering 2018 - 2021

Bhagalpur College of Engineering, Bhagalpur · CGPA: 8.75/10.0

Advisor: Prof. Arjun Kumar

Diploma in Electrical Engineering 2015 - 2018

Government Polytechnic Muzaffarpur, Muzaffarpur · 77.73%

Achievements

AIR 221 · GATE EE 2022; AIR 227 · GATE IN 2022; Rank 1 · BCECE LE 2018.